

Number System

For any doubt, check Video →



NS Revision

TOPIC 1 → Integral Solving variety

(V) $x^2 - y^2 = K$

(C1) $x^2 - y^2 = (\text{multiple of } 4)$
 but not 5
 Ex $x^2 - y^2 = 40$

(S1) $40/4 = 10$

(S2) Fac of 10 → 4 +

(S3) +ve = $\frac{\text{Fac}}{2} \rightarrow 2$

(S4) TS = $4 \times 2 = 8$

(C2) $x^2 - y^2 = (4k)$
 but 15
 $x^2 - y^2 = 36$

(S1) $36/4 \rightarrow 9$

(S2) Fac of 9 → 3

(S3) +ve = $\frac{\text{Fac} - 1}{2} = \frac{3-1}{2} = 1$

(S4) TS = $4 \times 1 + 2 \rightarrow 6$

(C3) $x^2 - y^2 = 4K + 2$
 0+

(C4) $x^2 - y^2 = \text{odd} (\text{not } 15)$
 $x^2 - y^2 = 21$

(S1) Fac of 21 → 4

(S2) +ve → $\frac{4-1}{2} = 2$

(S3) TS → $4 \times 2 = 8$

(C5) $x^2 - y^2 = \text{odd} + 15$
 $x^2 - y^2 = 81$

(S1) Fac of 81 → 5

(S2) +ve → $\frac{5-1}{2} = 2$

(S3) TS = $4 \times 2 + 2 = 10$

Bhool gaye?
 chalo Pousi NS videos
 Revise Karo =



Topic 2 - where $x \neq y$ x, y are natural no

$$x^2 - y^2 = 20$$

You have to solve this

$$\begin{aligned}(x-y)(x+y) &= 20 \\ &= 1 \times 20 \\ &= 2 \times 10 \\ &= 4 \times 5 \\ &= 5 \times 4 \\ &= 10 \times 2 \\ &= 20 \times 1\end{aligned}$$

$x-y=2$	$x-y=10$
$x+y=10$	$x+y=2$
$2x=12$	
$x=6$	$x=6$
$y=4$	$y=-4$

Topic 3 - ~~where~~

$$a \times b = 20$$

a, b Natural no

$$\begin{aligned}&= 1 \times 20 \\ &= 2 \times 10 \\ &= 4 \times 5 \\ &= 5 \times 4 \\ &= 10 \times 2 \\ &= 20 \times 1\end{aligned}$$

Pair will be factor of 20 $\rightarrow$ 6

$$\begin{aligned}a > b &\rightarrow 3 \\ a < b &\rightarrow 3\end{aligned}$$

$$a \times b = 36$$

Pair = factor of 36 $\rightarrow$ 9

$$\begin{aligned}a > b &\rightarrow 4 \\ a < b &\rightarrow 4 \\ a = b &\rightarrow 1\end{aligned}$$

Topic 4

$$ab+bc=19$$

$$bc+ca=28$$

Find abc (min or max)

$$b(a+c)=19$$

$$(b=1, a+c=19)$$

$$C(a+b)=28$$

$$=1 \times 28$$

$$=2 \times 14$$

$$=4 \times 7$$

$$C=1, a+b=28$$

$$C=2, a+b=14$$

$$C=4, a+b=7$$

(a, b) (Natural
vals)

Topic 5

$$\frac{a^2}{2} + ab + a = 5$$

$$b + ab + b = 7$$

Find $a+b$

Qus will revolve
around addition
or subtraction
of these two
eqn.

Topic 6

$$3x+5y=20$$

Find Integral Soln $x \geq y$
+ve

$$\text{Solve } 5y = 20 - 3x$$

$$y = 4 - \frac{3x}{5}$$

$$x=5, y=1$$

$$x=10, y=-2$$

$$x=15, y=-5$$

Add on eqn

$$x \leq 30$$

$$y \leq 0$$

Solving $\rightarrow$ add $\rightarrow a^2 + b^2 + 2ab + a + b = 12$
 $(a+b)^2 + (a+b) = 12$
 $a+b = x$
 $x^2 + x - 12 = 0$

$$3x + 5y = 20 \quad (\text{Natural})$$

$$y = 4 - \frac{3x}{5}$$

$$\text{SO } x=5, y=1$$

Now x, y ke coeff sath Badega
 $5y, x$ ke coeff sath.

Blood game?
 check
 common terms
 in Algebra
 or check
 LCM 500
 Complete
 Series.

Topic 7

$$\frac{a}{x} + \frac{b}{y} = \frac{1}{k}$$

+ve Solution $\leftrightarrow$ Factors of $a \times b \times k^2$
 Total Sol $\leftrightarrow$ (2x +ve Solution) - 1

$$\text{Ex } \frac{2}{x} + \frac{3}{y} = \frac{1}{5}$$

+ve $\leftrightarrow$ Factors of $2 \times 3 \times 5^2$

$\leftarrow (12)$

Total Solution $\leftrightarrow 2 \times 12 - 1 \leftrightarrow 23$ ✓

$$\frac{2}{x} + \frac{3}{y} = \frac{1}{5}$$

$$(x-10)(y-15) = 2 \times 3 \times 5^2$$

$$\textcircled{6} \rightarrow \frac{3}{x} + \frac{4}{y} = \frac{1}{7}$$

Find Total Solution

Topic 7

$$\frac{a}{x} + \frac{b}{y} = \frac{1}{k}$$

+ve Solution → Factors of $0 < x < y < k^2$

Total Sol → (2x +ve Solution) - 1

Ex $\frac{2}{x} + \frac{3}{y} = \frac{1}{5}$

+ve → Factors of $2 \times 3 \times 5^2$

→ (12)

Total Solution → $2 \times 12 - 1 \rightarrow 23$ ✓

$$\frac{2}{x} + \frac{3}{y} = \frac{1}{5}$$

$$(x-10)(y-15) = 2 \times 3 \times 5^2$$

③

$$\frac{3}{x} + \frac{4}{y} = \frac{1}{7}$$

Find Total Solution

+ve → Fac of $3 \times 4 \times 7^2$

$$\rightarrow 2 \times 3 \times 3 \rightarrow 18$$

$$\text{Total Sol} \rightarrow 2 \times 18 - 1 \rightarrow 35$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{36}$$

+ve Sol → Factors of $1 \times 1 \times (6^2)^2$

$$\rightarrow 6^4$$

$$\rightarrow 2^4 \times 3^4$$

+ve Sol → 25

$$\text{Total Sol} \rightarrow 2 \times 25 - 1 \rightarrow 49$$

(+ve)

$$\begin{aligned} x &< y \rightarrow 12 \\ x &< y \rightarrow 12 \\ x &= y \rightarrow 1 \end{aligned}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{5}$$

(+ve)

$$\rightarrow 1 \times 1 \times 5^2 \rightarrow 3$$

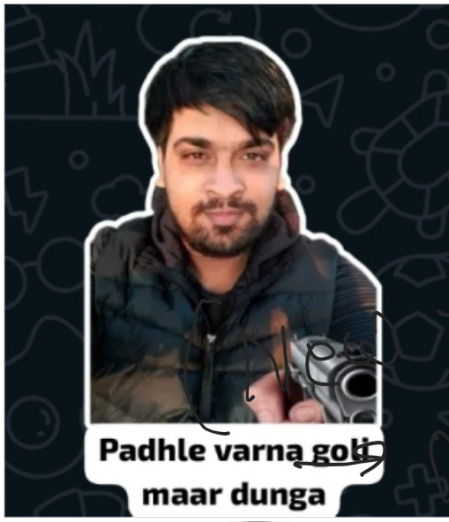
$$x=1, y=5$$

$$x=5, y=1$$

$$x=5, y=5$$

NIS udan concept 1

Kam do cover
Cat ke baad karoge!



Topic to learn
(new video aagaya)



Topic 8

(C1) $|x|=2$ → Total 2

(C2) $|x|+|y|=2$ ✓
8 Solutions
 $4n \rightarrow 4 \times 2 \rightarrow 8$
Ex $|x|+|y|=3$ +
 $4 \times 3 = 12$
 $|x|+|y|=2$

$x \geq 0, y \geq 0$
 $x+y=2$
 $x=0, y=2$
 $x=1, y=1$
 $x=2, y=0$

$x \geq 0, y \leq 0$
 $x-y=2$
 $x=y+2$
 $y=0, x=2$
 $y=-1, x=1$
 $y=-2, x=0$
 ~~$y=-3, x=-1$~~

$x \leq 0, y \geq 0$ ✓
 $-x+y=2$
 $-x=y+2$
 $y=0, x=-2$
 $y=1, x=-1$
 $y=2, x=0$

$x \leq 0, y \leq 0$
 $-x-y=2$
 $-x=y+2$
 $y=0, x=-2$
 $y=-1, x=-1$
 $y=-2, x=0$

$|x|+|y|=6$ ✓

imp method / mostly used

$\hookleftarrow \hookleftarrow \hookleftarrow$
(C4) $|x|+|y|+|z|=n$
 $4n^2+2$
 $|x|+|y|+|z|=1$



check QA(500) series
video 2

6th

#

$$7-2$$

$$(x) + (y) + y = 5$$

(find x, y)

$$(2x) + y + (y) = 7$$

4 cases

(1) $x > 0, y > 0$

$$2x + y = 5$$

$$2x + 2y = 7$$

$$y = 2$$

$$x = 3/2$$

(2) $x > 0, y \leq 0$

$$2x + y = 5$$

$$2x = 7$$

$$x = 7/2$$

$$y = -2$$

(3) $x \leq 0, y > 0$

$$y = 5$$

$$2x + 2y = 7$$

$$2x = -3$$

$$x = -3/2$$

(4) $x \leq 0, y \leq 0$

$$y = 5$$

$$x = 7/2$$

EK Baar x +ve, y +ve
 x +ve, y -ve
 x -ve, y -ve
 x -ve, y +ve

(Algebra Topic Test main
seekha tha)



X

$$\begin{aligned} (x) + (x) + y &= 5 \\ (2x) + y + (y) &= 7 \end{aligned} \quad (\text{find } x, y)$$

4 cases

(1) $x > 0, y > 0$

$$\begin{aligned} 2x + y &= 5 \\ 2x + 2y &= 7 \end{aligned}$$

$$\begin{aligned} y &= 2 \\ x &= 3/2 \end{aligned}$$

(2) $x > 0, y \leq 0$

$$\begin{aligned} 2x + y &= 5 \\ 2x &= 7 \\ x &= 7/2 \\ y &= -2 \end{aligned}$$

(3) $x \leq 0, y > 0$

$$\begin{aligned} y &= 5 \\ 2x + 2y &= 7 \\ 2x &= -3 \\ x &= -3/2 \end{aligned}$$

(4) $x \leq 0, y \leq 0$

$$\begin{aligned} y &= 5 \\ x &= 7/2 \end{aligned}$$



Topic 10

$$A^B = L$$

- 1) $A = 1$, $B = \text{any value}$
- 2) $A = -1$, $B = \text{even}$
- 3) $B = 0$, $A = \text{any value } A \neq 0$

Ex

$$(x^2 - 3x + 3) = 1$$

$$1) B=0 \quad (x=1)$$

$$2) x^2 - 3x + 3 = 1$$

$$x^2 - 3x + 2 = 0$$

$$x = 1, 2$$

$$3) x^2 - 3x + 3 = -1$$

$$\cancel{x^2 - 3x + 4 = 0}$$

$$x =$$

Topic 11

$$A^B = (64) \quad (A \neq B \text{ are integers})$$

$$A^B = (64)^1$$

$$= (2^6)^1 = (2^3)^2 = (2^2)^3$$

$$\rightarrow 64 = \begin{pmatrix} 8^2 \\ (-8)^2 \end{pmatrix} (4^3)$$

$$(2)^6 \leftrightarrow (-2)^6$$

$$(64)^1$$

$$(2)^6 \quad (-2)^6$$

$$(4)^3$$

$$(8)^2 \quad (-8)^2$$

$$64$$

check vdaan concept (Algebra)

$N \pmod{3} = 1$
 $\pmod{4} = 2$
 $\pmod{5} = 3$
 $N = K \text{LCM}(3, 4, 5) - 2$
 $N = K \text{LCM}(60) - 2$
 $N = 60K - 2$ ✓
 2 N is all div by 7
 $N = 7a$ ✓
 $7a = 60K - 2$
 $a = \frac{60K - 2}{7}$
 $= 50K + \frac{4K - 2}{7}$
 $= \frac{50K + 2}{7} \left[\frac{2K - 1}{7} \right]$
 $\begin{cases} K=4, \\ K=11 \end{cases}$

$N \pmod{3} = 1$
 $\pmod{4} = 1$
 $\pmod{5} = 1$
 $N = K \text{LCM}(3, 4, 5) + 1$
 $N = 60K + 1$
 3 digit $60K + 1$ > 100
 $60K > 99$
 $K > 1.65$
 $K = 2$

2 digit $60K + 1$
 $60K + 1 \leq 99$
 $60K \leq 98$
 $K \leq 1.63$
 $K = 1$
 121
 181
 241
 961
 $\frac{121 + 961}{2} = \frac{1082}{2} = 541$

Successive division
 $N \text{ Success } 3, 4, 5$
 $r = 1, 2, 3$
 $N = 3a + 1$ — (1)
 $a = 4b + 2$ — (2)
 $b = 5c + 3$ — (3)
 $a = 4 \times (5c + 3) + 2$
 $a = 20c + 14$
 $N = 3(20c + 14) + 1$
 $N = 60c + 43$

Successive ko Dhyaan se
 Dekho $\rightarrow$ Success wahi
 se milegi



1) $a \times b = 50$

unorder $\rightarrow 2^{p-1} \rightarrow 2^{2-1} \rightarrow 2$

nod pair

2) $a + b = 50$

unorder $\rightarrow E(50) \rightarrow 50 \times \frac{1 \times 4}{2} \rightarrow 100 \rightarrow 10$

ord $\rightarrow 2 \times 10 = 20$

Two no (A, B) their Hcf is 5 & Sum is 100.
Find no of pair.

$5a, 5b$

$5a + 5b = 100$

$a + b = 20$

ord $\rightarrow E(20) = 20 \times \frac{1 \times 4}{2} = 40$

$\rightarrow 1$

$Hcf \times LCM = 1^{st} N \times 2^{nd} N$

2 no ✓

"Ye To Pakka Bhool gaye
Hoge. Jao HCF LCM
Revise Karo."



Topic 121

Divisibility Rule

div by 2 → Last digit

div by 3 → Sum of digits

div by 4 → Last 2 digits

divisor 10^n Factor

$25 \rightarrow 10^2$

$125 \rightarrow 10^3$

div Rule of $(10^n + 1)$ Fun div

div Rule of $1001 \rightarrow 10^3 + 1$

$123456789 \pmod{1001}$

$789 - 456 + 123 \pmod{1001}$

$1001 \rightarrow 7 \times 11 \times 13$

$(10^2 + 1) \rightarrow 101$

$10^1 + 1 \rightarrow 11$

div Rule of $(10^3 - 1)$

$27 \times 37 = 999 \rightarrow 10^3 - 1$

$10^3 - 1$

$123456789 \pmod{10^3 - 1}$

$789 + 456 + 123$

$4 \rightarrow 2$

$2 \rightarrow 4$

Remainder

✓ $\begin{cases} x^n - y^n \text{ will be div by } (x-y \text{ \& } x+y) \text{ if } n \text{ is even} \\ \longrightarrow (x-y) \text{ if } n \text{ is odd} \\ x^n + y^n \longrightarrow (x+y) \text{ if } n \text{ is odd} \end{cases}$

Ex ✓

$$\begin{aligned} 7^6 - 6^6 &\leftrightarrow 3, 11 \\ (7^2)^3 - (6^2)^3 &\rightarrow 13 \\ (7^3)^2 - (6^3)^2 &\leftrightarrow 127 \\ &\quad 7^3 + 6^3 \leftarrow 519 \end{aligned}$$

Ex $\leftrightarrow 7^9 + 8^9 + 9^9 + 10^9 \rightarrow 3491$

Imp

$$\begin{aligned} N \bmod 2 &= 0 \\ N \bmod 4 &\neq 0 \end{aligned}$$

$N \bmod \text{LCM}(2, 4)$

$N \bmod 4 = 0$ ✓

Negative

$$5^{10} \bmod 9$$
$$(-1)^{10}$$
$$1$$

$$7 \bmod 100 \rightarrow 1$$

$$7 \rightarrow 1$$

$$E(2) \bmod 1 = 1$$

Copain

$$42 \bmod 100$$

$$1) E(100) = 40$$

$$2) 42 \bmod 40 = 2$$

$$3) 7^2 \bmod 100 = 49$$

$$E(100) = 100 \times \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{7}\right)$$

$$= 100 \times \frac{1}{2} \times \frac{6}{7} \rightarrow 40$$

Wilson

$$\begin{aligned} p-1 \bmod p &= p-1 \\ p-2 \bmod p &= 1 \\ p-3 \bmod p &= \left(\frac{p-1}{2}\right) \end{aligned}$$

$$\begin{aligned} 1 \times 12 + 2 \times 12 + 3 \times 12 \\ 1 \times 12 &= 12 - 12 \\ n \times 12 &= 12n - 12n \\ 1 \times 12 + 2 \times 12 + 3 \times 12 + \dots + 15 \times 12 \bmod 17 \\ 12 - 12 + 24 - 12 + \dots - 12 + 12 \\ 12 &= -1 \bmod 17 \\ 16 - 1 &\rightarrow 15 \end{aligned}$$

Ex

$$\begin{aligned} 16 \bmod 17 &= 16 \\ 15 \bmod 17 &= 1 \\ 14 \bmod 17 &= 8 \\ 13 \bmod 17 &= 3 \end{aligned}$$

$$\begin{aligned} 15 \bmod 17 &= 1 \\ 15 \times 14 \times 13 \bmod 17 &= 1 \\ 6R &= 17Q + 1 \end{aligned}$$

$$\begin{aligned} Q &= 1 \\ R &= 3 \end{aligned}$$

$$16 \bmod 17 = 0$$

V.I.P. $\Rightarrow$ Power in Factorials / Towering
Factors

Point
$20 \rightarrow 2$
 $\frac{20}{2} = 10$
 $\frac{10}{2} = 5$
 $\frac{5}{2} = 2$
 $\frac{2}{2} = 1$
 $\frac{1}{2} = 0$
 2^8

$20 \rightarrow 5$
 $\frac{20}{5} = 4$
 $\frac{4}{5} = 0$

$3 \rightarrow 4 \leftarrow 2^2$

$20 \rightarrow 2^8 \leftarrow (2^2)^{4/2} \rightarrow 4^2$

$20 \rightarrow 9$

$\frac{20}{3} = 6$ $3^8 \leftarrow 9^4$
 $\frac{6}{3} = 2$

$20 \rightarrow 6$ 2
 3

$20 \rightarrow 2^8$ 8
 $20 \rightarrow 3^8$ 8

$2^8 \times 3^8 \times 2^8 \times 2^{10}$
 $2 \rightarrow 4 \leftarrow 4^9 \leftarrow 12^8$
 $3 \rightarrow 3^8$

No of trailing zeros $\rightarrow (10) \rightarrow 2 \times 5$

$$(120) \leftrightarrow 2^{18} \times 5^4$$

$$(120)^{30} \leftrightarrow (5^4)^{30} \leftrightarrow 5^{120}$$

$$51 \rightarrow x$$

$$55 \rightarrow x+1$$

$$99 \rightarrow y$$

$$100 \rightarrow y+2$$

$$100 \times 9$$

$$12 \rightarrow z$$

$$(125) \leftrightarrow (z+3)$$

Reverse

n has 10 trailing zeros

$$10 \times 4 = 40$$

$$40 \leftrightarrow 40/5 = 8$$

$$9$$

$$45 \rightarrow 10$$

$$45 - 41$$

$$50$$

$$2 \times 5 = 10$$

Product of factor $\leftarrow (N)^{\frac{\text{Fact}}{2}}$

Fact inc/dec order

$$10 = 1, 2, 5, 10$$



$$16 \rightarrow (1), 2, (4), 8, (16)$$



$$N = PS \leftarrow \text{Factor} \leftarrow \text{odd}$$

IF Product of 10th Fact $\times$ 13th Fact $= N$

Find Total no of fac

$$10\text{th Fac} \times 13\text{th} = N$$

$$9\text{th Fac} \times 14\text{th} = N$$

...

$$1\text{st Fac} \times 2\text{nd} = N$$

$$a\text{th Fac} \times b\text{th Fac} = N$$

$$(a+b-1)$$

$$10+13-1 \rightarrow 22$$

$$N = 100 = 2^2 \times 5^2 \leftarrow TF = 9$$

middle factor $\leftarrow$

$$\sqrt{N} = \sqrt{100} \Rightarrow 10^+$$

Sum of fact

$$N = 2^2 \times 3^3 \times 5^2$$

$$(\cancel{2^0} + 2^1 + 2^2) \times (3^0 + 3^1 + 3^2 + 3^3) \times (\cancel{5^0} + 5^1 + 5^2)$$

Sum of fact that are div by 10 $\left\{ \begin{array}{l} 2 \\ 5 \end{array} \right.$

N has got 8 factors
Find No of possible factors of N^2

$$\begin{aligned} N = a^7 &\rightarrow N^2 = a^{14} \rightarrow 15 \\ N = a^2 b^3 &\rightarrow N^2 = a^4 b^6 \rightarrow 21 \\ N = abc &\rightarrow N^2 = a^2 b^2 c^2 \rightarrow 27 \end{aligned}$$

- a) 15
- ~~b) 17~~
- c) 21
- d) 27

17 not possible



N has 5 Factors

Find all 3 digit possible no

$$N = a^4$$

$$2^4 = 16$$

$$3^4 = 81$$

$$5^4 = 625$$

$$7^4 = 2401$$

N has 200 Factors

min N

$$200 = 2 \times 2 \times 2 \times 5 \times 5$$

$$N = a^1 \times b^1 \times c^1 \times d^4 \times e^4$$

Common Factors

$3 \times 4 \times 4 \times 4 = 48 \rightarrow N_1 = 2^2 \times 3^3 \times 5^3$
 $72 \rightarrow N_2 = 2^3 \times 3^2 \times 5^1$

HCF = $2^2 \times 3^2 \times 5^1$
 $\Rightarrow 3 \times 3 \times 4 \rightarrow 36$ ✓

Find N_1 that are not common with N_2

$48 - 36 \rightarrow 12$ ✓

Find N_2 that are not common with N_1

$72 - 36 = 36$

Factor $N = 2^3 \times 3^5 \times 10^4$
 $= 2^3 \times 3^5 \times 2^4 \times 5^4 \rightarrow 2^7 \times 3^5 \times 5^4$

2401

NS - Factor

① $N = 2^2 \times 3^2 \times 5^3$

TF = $4 \times 3 \times 4 \rightarrow 48$

Even $\leftarrow$ (Ignore 2^0) $\rightarrow 3 \times 3 \times 4 \rightarrow 36$

odd $\leftarrow$ (Take only 2^0) $\rightarrow 1 \times 3 \times 4 \rightarrow 12$

PS $\leftarrow (2^0, 2^1) (3^0, 3^1, 3^2) (5^0, 5^1, 5^2, 5^3)$

$2 \times 2 \times 2 \rightarrow 8$

PS other than 1 $\leftarrow 8 + 1 = 7$

(PC) $\rightarrow (2^0, 2^1) (3^0) (5^0, 5^1, 5^2, 5^3)$

$2 \times 1 \times 2 \rightarrow 4$

PS or PC

PS + PC = (PS/PC)

$a^2 + a^3 = (a^6)$

NS

all imp points (summary)

div by 24 $\leftarrow 0/53 \rightarrow (2^3, 3^1)$
 $\rightarrow (2^3) (3^1, 3^2) (5^0, 5^1, 5^2, 5^3)$
 $1 \times 2 \times 4 \rightarrow 8$

Common Fac

$N_1 =$ (Take HCF & find fac of hcf)

$N_2 =$

Prod of fact $\leftarrow (N) \frac{\text{no of fac}}{2}$

Sum of Fact

~~for~~ $N = 2^2 \times 3^2 \times 5^3$

$(2^0 + 2^1 + 2^2) (3^0 + 3^1 + 3^2) \times (5^0 + 5^1 + 5^2 + 5^3)$

Middle Factor = $\sqrt{N}$ (In case of odd no of Fac)

IF both Fac $\times$ Mth Fac = N

Total Fac = $10 + 14 - 1 = 23$

$N \rightarrow$ Factor $\rightarrow 10$

$10 = 10 = 2 \times 5$

$N = a^2 \leftarrow$

$N = a^2 b^4$

Q98 NS

HCF, LCM $\leftrightarrow N = D \times a + R$

last concept of ns

$$N \xrightarrow{D} 15 \xrightarrow{R} 12$$

$$N = 15a + 12$$

Find Rem when same N is divided by 5.

$$N = D \times a + 15$$

$$2N = 1 \times b + 12$$

$$2N = 2D \times a + 30$$

$$0 = D(2a - b) + 15$$

$$D(b - 2a) = 15$$

$$D = 1, 2, 3, 6, 15$$

Find possible value of D

$$D = 15$$

Imp Hai - Bhoolna mat

Find Largest Number that when divide 100, 130, 160 leaves Some Remainder.

$$\begin{aligned} 100 &= D \times a + R \quad (1) \\ 130 &= D \times b + R \quad (2) \\ 160 &= D \times c + R \quad (3) \end{aligned}$$

$$\begin{aligned} (1) - (2) &= 30 \\ (2) - (3) &= 30 \\ (3) - (1) &= 60 \end{aligned}$$

$$\text{HCF of } (30, 30, 60)$$

$$\Rightarrow 30$$

Find Largest Number that when divide 100, 130, 160 leave same rem

$$\begin{aligned} 100 &= D \times a + 5 \\ 130 &= D \times b + 5 \\ 160 &= D \times c + 5 \end{aligned}$$

$$\begin{aligned} (1) - (2) &= 95 \\ (2) - (3) &= 125 \\ (3) - (1) &= 155 \end{aligned}$$

$$D = \text{Factor}(95, 125, 155)$$

$$5$$

ALL THE BEST